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**Project:** Measuring the effect of stimulus position and order of presentation on observer choice in a visual psychophysics task

**Introduction:** Observer's decision in a two-alternative/two-interval forced choice experiment depends on the order in which stimuli are presented. We aim to assess the magnitude of this dependence by measuring the psychometric functions of human observers in a two-alternative/two-interval forced-choice luminance discrimination task.

Observers will compare two achromatic square patches to determine whether the two squares are same or different in luminance/brightness. One patch (the standard) will remain fixed in luminance, while the other (the comparison) will vary in its luminance. We will measure the psychometric function for the same-different discrimination task, i.e., the relationship between comparison patch luminance and the fraction of times it is chosen to be the same as the standard patch.

We will perform two experiments. In one experiment the standard and comparison stimuli will be presented one after the other at the same location in a two-interval forced choice task (Figure 1). In the other experiment, the standard and comparison stimuli will be presented side-by-side in a two-alternative forced choice task (Figure 2). In both experiments, the order of the standard and comparison stimuli will be randomized across trials and we will study whether observer's choice varies in similar ways with the interval (first or second) or the side (right or left) on which standard stimuli is presented.

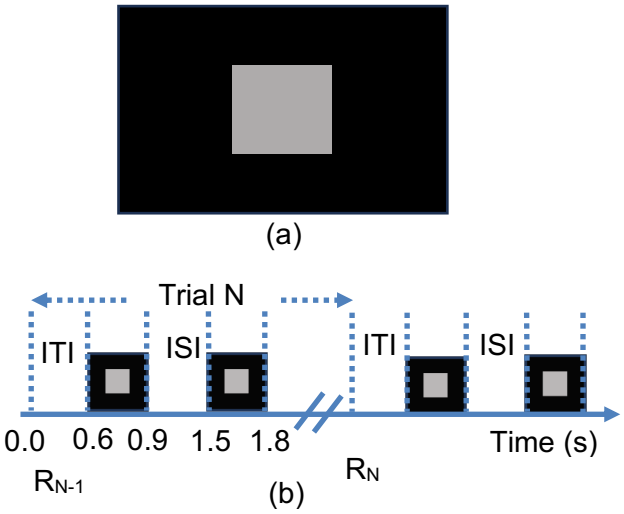



Figure 1: (a) An example of stimulus presented in the temporal order effect (2IFC) experiment. The gray patch represents the stimulus. The black background represents rest of the monitor. (b) The sequence of a typical trial. The response $R_{(N-1)}$ indicates the end of the (N-1)th trial. The Nth trial begins 0.6s after the (N-1)th response (Inter Trial Interval, ITI). The Nth trial consists of two 0.3s stimuli intervals with a 0.6s inter-stimulus interval (ISI) in between. The observer will respond after the second stimulus is removed from the screen by pressing a button on a gamepad. The observer can take as long as they wish before making the response ($R_N$). The next trial begins 0.6s (ITI) after the response.

**Methods**: We will perform the following two experiments.

**Sequential presentations (2IFC):** In this experiment, we will display two square patches sequentially one after the other on a color-calibrated monitor. The squares will be 1.68 degrees in visual angle and will be shown on an otherwise dark monitor. Each panel will be presented for 0.3s with an inter-stimuli interval of 0.6s. Observers will indicate their response after both stimuli have been presented and removed from the screen by pressing on a gamepad. Observer can take as long as they wish to make their decision. The next trial will begin 0.6

seconds after their decision has been recorded (inter-trial interval, ITI). The actual inter-trial interval will be longer and variable.

**Simultaneous presentations (2AFC):** In this experiment, both the standard and comparison square patches will be presented simultaneously side-by-side. The size of the square patches will be the same as the sequential effect experiment (1.68 degrees in visual angle). The patches will be 5.57 degrees visual angle apart. Observers will view the stimuli for as long as they wish before making their decision. Observers will indicate whether they perceive the patches to be the same or different by pressing on a gamepad. Once the decision is recorded, the screen will turn black and the next trial will begin after a delay of 0.6s (inter-trial interval, ITI). Thus, the actual trial duration will be variable.

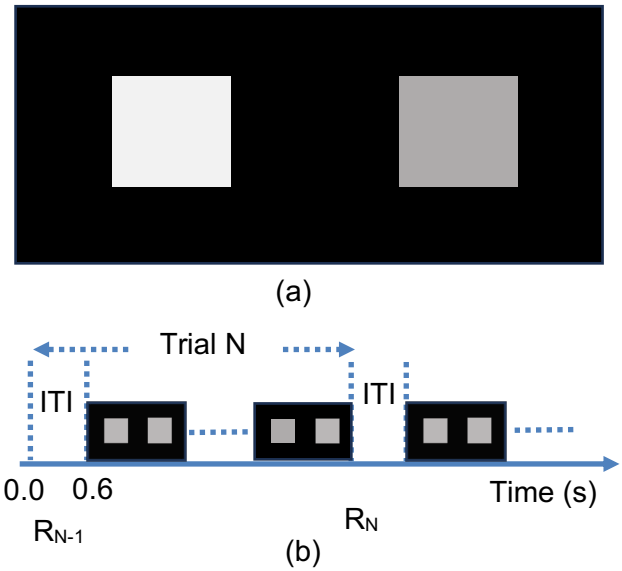



Figure 2: (a) An example of stimulus presented in the space order effect (2AFC experiment). The observers will view two achromatic square patches presented on an otherwise dark background and will report if the patches appear same or different. In this figure the two patches are different. (b) The Nth trial starts 0.6s (ITI) after the recording of the $R_{(N-1)}$ response. The stimulus will be displayed throughout the trial till the observer makes their decision. Observers will not be asked to fixate.

**Common settings to both experiments**: We will measure psychometric functions for 3 values of standard square patch luminance. The experiments will be structured as follows:

The recording of data at one value of standard square luminance in an experiment will be called one block. Thus, we will collect data for three blocks for each experiment. Each block will consist of 336 trials that will consist of three sub-blocks of 112 trials. The observer will take a minimum rest of 1 minute after each sub-block. The observer may take additional breaks as needed.

For each block, we will collect same-different response at 7 values of comparison square luminance placed symmetrically around the standard square luminance in logarithmic steps. We will perform 48 trials at each value of the comparison square luminance. In temporal order effect experiment, 24 of these trials will have standard stimulus presented on the first interval and the other 24 trials on the second interval. In the space order effect experiment, the standard will be on the left in 24 trials and on the right in other 24 trials. The interval/position and luminance of the comparison square will be randomized over trials.

Both experiments will have identical parameters for the luminance of the standard, comparison, and background luminance. The values of the parameters are given in the Table below (Table 1):

| Block 1<br>Standard RGB = (30,30,30), 13.2 cd/m$^2$ | | | | | | | |
|---|---|---|---|---|---|---|---|
| Comparison (R,G,B) | (20,20,20) | (23,23,23) | (26,26,26) | (30,30,30) | (34,34,34) | (40,40,40) | (45,45,45) |
| CIE-Y (cd/m$^2$) | 8.8 | 10.1 | 11.4 | 13.2 | 14.9 | 17.6 | 19.8 |
| Block 2<br>Standard RGB = (60,60,60), 26.4 cd/m$^2$ | | | | | | | |
| Comparison (R,G,B) | (40,40,40) | (45,45,45) | (52,52,52) | (60,60,60) | (69,69,69) | (79,79,79) | (91,91,91) |
| CIE-Y (cd/m$^2$) | 17.6 | 19.8 | 22.4 | 26.4 | 30.3 | 34.7 | 39.96 |
| Block 3<br>Standard RGB = (90,90,90), 39.5 cd/m$^2$ | | | | | | | |
| Comparison (R,G,B) | (59,59,59) | (68,68,68) | (78,78,78) | (90,90,90) | (103,103,103) | (119,119,119) | (136,136,136) |
| CIE-Y (cd/m$^2$) | 25.9 | 29.9 | 34.3 | 39.5 | 45.2 | 52.3 | 59.7 |

*Table 1: Value of standard and comparison patches used in the experiments.*

To familiarize the observer with the experiments, at the beginning of each experiment a Block 0 will be conducted. The parameters of Block 0 will be identical to Block 2.

The data for space order effect experiment will be collected in one session with all four blocks conducted during the session. Due to the extended nature of the temporal order effect, the data for this experiment will be collected in two sessions conducted on separate days. The first session will consist of Block 0 and another block chosen randomly from block 1, 2 and 3. The second session will have other two blocks chosen in a random order. Thus, each observer will participate in three sessions (one for space order effect and two for temporal order effect). These three sessions will be conducted on three separate days.

We will collect data for 20 naïve observers. The order of the blocks will be counterbalanced across observers before data collection. Observers who do not complete all three blocks (for example, by not showing up for scheduled sessions) will be excluded and replaced with a new subject, and their data will be discarded.

The order of the experiment (space or temporal) will be chosen randomly for each observer. For each observer, all the data for their first experiment will be collected before starting the other experiment.

**Observer Recruitment and Exclusion:** We will recruit observers from Haverford College and the local community. These observers will be screened to have normal (or corrected-to-normal) visual acuity (20/25 or better) and normal color vision (as tested with pseudo-isochromatic plates). The observers will receive $15/session for their participation. The data for each observer will be collected over three experimental sessions.

The subjects will be familiarized with both experiments and the experimental set up before data collection. For this, we will run a familiarization block with 84 trials with the parameters of Block 2, but only 12 repetitions per comparison luminance level.

**Stimuli Generation and Presentation:** Stimuli will be generated using the Unity gaming engine and will be presented on an LCD display. The monitor will be color calibrated before data collection. The monitor will be located at a distance of 120 cm from the observer. The size of each square patch will be 3.5cm x 3.5cm on the monitor, corresponding to 1.66x1.66 degrees of visual angle. The center of the two squares will be separated by 11.7cm. This corresponds to 5.57 degrees visual angle.

In the sequential effect experiment, each stimulus will be presented for 0.3 s, with an inter-trial interval of 0.6s. The response for each trial will be collected after the stimuli has been displayed and removed from the screen. The observer can take as long as they wish before entering the response. The next trial will start 0.6s (inter-trial interval) after the recording of observer's response.

In the space order effect experiment, the stimulus will be displayed from the beginning of the trial till the observer makes their decision. The next trial will start 0.6s (inter-trial-interval) after the recording of observer's response.

**Data-Analysis:** Data will be analyzed separately for each observer. We will fit a difference of Gaussians to the fraction same/different data using the maximum likelihood methods separately for comparison stimuli presented on first/second interval and left/right position. Specifically, for 2IFC data, we will fit the three-parameter functions

$$\Upsilon_{CF}(x) = \Phi\left(\beta_{CF}(x-\alpha_{1;CF})\right) - \Phi\left(\beta_{CF}(x-\alpha_{2;CF})\right) \quad (1a)$$

$$\Upsilon_{CS}(x) = \Phi\left(\beta_{CS}(x-\alpha_{1;CS})\right) - \Phi\left(\beta_{CS}(x-\alpha_{2;CS})\right), \quad (1b)$$

where Φ is the unit-normal cumulative Gaussian distribution, subscripts CF and CS stand for "comparison first" and "comparison second", and *x* is the comparison luminance. For 2AFC data, we will fit analogous functions in which subscripts CF and CS are respectively replaced with CL (for "comparison left") and CR (for "comparison right").

Type A effects have been defined as differences in the location of psychometric functions for each presentation order/position. Type B effects have been defined as differences in the slope of psychometric functions for each presentation order/position. The sign and magnitude of Type A effects are given by the sign and absolute value of $\frac{2(\alpha_{CF}-\alpha_{CS})}{1/\beta_{CF}+1/\beta_{CS}}$ (for 2IFC data and, with a straightforward change of subscripts also for 2AFC data), where $\alpha_{CF}=\frac{\alpha_{1;CF}+\alpha_{2;CF}}{2}$ and $\alpha_{CS}=\frac{\alpha_{1;CS}+\alpha_{2;CS}}{2}$ (and analogously with a change of subscripts for 2AFC data). The sign and magnitude of Type B effects are given by the sign and value of $\log_2(\beta_{CF}/\beta_{CS})$ for 2IFC data and analogously with a change of subscripts for 2AFC data.

Our goal is to determine whether Type A and Type B effects have similar sign and magnitude in temporal 2IFC and in spatial 2AFC. With respect to sign and separately for each type of effect, we will test the null statistical hypothesis that the prevalence of negative effects is the same in temporal 2IFC and in spatial 2AFC, using a test for equality of two proportions with independent samples. With respect to magnitude, and also separately for each type of effect,

we will characterize the distribution of effects and compare them with a statistical test for homogeneity of distributions.

Secondary to our main goal, we will also assess the extent to which a difference limen computed from aggregated data across presentation orders/positions (which is the conventional approach) differs from difference limens computed separately from data for each presentation order/position.